Absolutely. Based on your uploaded material, the topic is **Integration**, and I'll turn it into **80/20 interview-focused notes** rather than reproducing all of the rigorous proof-heavy material.

The source emphasizes integration as the inverse of differentiation, the area-under-the-curve interpretation, the Fundamental Theorem of Calculus, integration techniques, and numerical integration such as the trapezoidal rule.

Integration — 80/20 Interview Notes

1. Integration — ⭐⭐⭐⭐⭐

Definition

Integration is the process of accumulating small quantities to obtain a total quantity.

Geometrically, a definite integral represents the **signed area under a curve**:

$$\boxed{\int_a^b f(x)\,dx}$$

Think:

Differentiation → rate of change
Integration → accumulation / total

The source introduces integration by asking the reverse question of differentiation: if velocity is known, can we reconstruct distance?

Simple intuition

Suppose velocity is:

$$v(t) = 5 \text{ m/s}$$

for 10 seconds.

Distance:

$$5 \times 10 = 50m$$

Graphically:

velocity

|

5 |————————————

| |

| |

0 +---------------+---- time

0 10

Area = width × height

= 10 × 5

= 50

Therefore:

$$\boxed{\text{Distance} = \int_0^{10} v\,(t)\,dt}$$

The uploaded source uses exactly this velocity-area interpretation.

---

2. Differentiation vs Integration — ⭐ ⭐ ⭐ ⭐ ⭐

This is an **extremely important interview concept**.

Differentiation	**Integration**
Measures rate of change	Measures accumulation
Breaks a quantity into local change	Combines small quantities into total
Slope	Area
Velocity from distance	Distance from velocity
$s(t) \rightarrow v(t)$	$v(t) \rightarrow s(t)$

Diagram

DIFFERENTIATION

↓

Distance s(t)

↓

Velocity v(t)

↓

Rate of change

INTEGRATION

↑

Distance s(t)

↑

Velocity v(t)

↑

Accumulation

The source explicitly describes integration as effectively reversing differentiation.

Interview answer

Differentiation tells us how quickly something is changing, while integration accumulates those small changes to determine the total quantity.

---

3. Indefinite Integral — ⭐ ⭐ ⭐ ⭐ ⭐

Definition

An **indefinite integral** finds a function whose derivative is the given function.

If:

$$F'(x) = f(x)$$

then:

$$\boxed{\int f(x)\,dx = F(x) + C}$$

where C is the constant of integration.

Example

Find:

$$\int 2\,x\,dx$$

We know:

$$\frac{d}{dx}(x^2) = 2x$$

Therefore:

$$\boxed{\int 2\,x\,dx = x^2 + C}$$

Why $+C$?

Because:

$$\frac{d}{dx}(x^2) = 2x$$

and

$$\frac{d}{dx}(x^2 + 5) = 2x$$

and

$$\frac{d}{dx}(x^2 - 100) = 2x$$

So there are infinitely many antiderivatives differing only by a constant. The source explicitly highlights this point.

---

4. Definite Integral — ⭐⭐⭐⭐⭐

Definition

A definite integral calculates accumulated quantity over a specific interval:

$$\boxed{\int_a^b f\,(x)\,dx}$$

Unlike an indefinite integral, it produces a **number**.

Example

$$\int_0^2 x\ dx$$

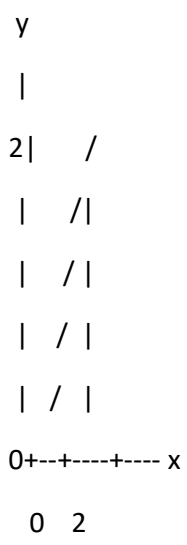
First find the antiderivative:

$$F(x) = \frac{x^2}{2}$$

Then:

$$\int_0^2 x\ dx = \left[\frac{x^2}{2}\right]_0^2$$
$$= \frac{4}{2} - 0$$
$$\boxed{= 2}$$

Diagram



Area = integral from 0 to 2

= 2

---

5. Signed Area — ⭐⭐⭐⭐

An integral represents **signed** area.

If the function is above the x-axis:

$$\int_a^b f(x)dx > 0$$

If below:

$$\int_a^b f(x)dx < 0$$

Example

$$f(x) = -2$$

from 0to 3:

$$\int_0^3 -2dx = -6$$

So the signed area is:

$$\boxed{-6}$$

The source explicitly distinguishes signed area from ordinary geometric area.

---

6. Fundamental Theorem of Calculus — ⭐⭐⭐⭐⭐

This is probably the **most important integration theorem for an interview**.

Formula

If:

$$F'(x) = f(x)$$

then:

$$\boxed{\int_a^b f\,(x)\,dx = F(b) - F(a)}$$

This is the **Newton–Leibniz formula**, also called the Fundamental Theorem of Calculus.

Example

Calculate:

$$\int_0^3 2\,x\,dx$$

Step 1 — Find antiderivative

$$F(x) = x^2$$

Step 2 — Apply theorem

$$F(3) - F(0)$$
$$= 9 - 0$$
$$\boxed{9}$$

Interview explanation

Instead of calculating infinitely many tiny areas directly, we find an antiderivative and evaluate it at the boundaries.

---

7. Why Does the Fundamental Theorem Work?

This is a good **follow-up interview question**.

Suppose:

$$F'(x) = f(x)$$

Then the integral accumulates all the tiny changes in F.

Conceptually:

Tiny changes:

F(x1)-F(x0)

+

F(x2)-F(x1)

+

F(x3)-F(x2)

+

...

+

F(xn)-F(xn-1)

Everything in the middle cancels.

= F(b)-F(a)

This is called **telescoping**.

The source's proof uses exactly this cancellation of intermediate terms.

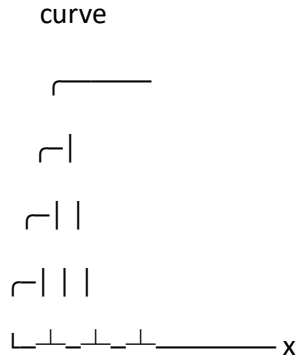
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8. Riemann Sum — ⭐ ⭐ ⭐ ⭐

This is the foundation behind numerical integration.

Idea

Instead of calculating the exact curved area directly, divide it into many small rectangles.



Each rectangle approximately has:

$$\text{Area} = \text{height} \times \text{width}$$

Therefore:

$$\boxed{\int_a^b f\,(x)dx \approx \sum_{i=1}^{n} f\,(x_i)\Delta x}$$

As the rectangles become extremely narrow:

$$\boxed{\int_a^b f\,(x)dx = \lim_{n\to\infty} \sum f\,(x_i)\Delta x}$$

The source develops integration from exactly this approximation idea.

---

9. Numerical Integration — ⭐ ⭐ ⭐ ⭐ ⭐

Why do we need it?

Sometimes finding an analytical antiderivative is difficult or impossible.

A famous example is:

$$\int e^{-x^2}\,dx$$

There is no elementary closed-form antiderivative.

The source contrasts this with differentiation: symbolic integration can be difficult, while numerical integration can still provide useful approximations.

Therefore:

Need integral

|

+---- Easy antiderivative?

| |

YES NO

| |

Analytical Numerical

integration integration

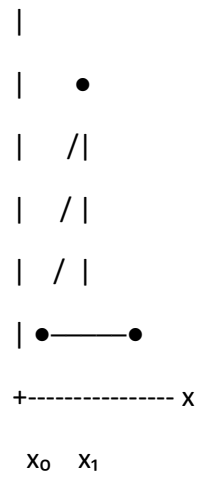
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10. Trapezoidal Rule — ⭐ ⭐ ⭐ ⭐ ⭐

This is one of the **highest-value numerical integration concepts** for an ML interview.

Instead of approximating the curve using rectangles, approximate it using **trapezoids**.

Diagram



Area of trapezoid:

$$A = \frac{1}{2}\big(f(x_0) + f(x_1)\big)(x_1 - x_0)$$

For many intervals:

$$\boxed{\int_a^b f\,(x)dx \approx \frac{b-a}{2n}\left[f(x_0) + f(x_n) + 2\sum_{i=1}^{n-1} f\,(x_i)\right]}$$

The source derives the trapezoidal rule by dividing the interval uniformly and summing trapezoid areas.

---

11. Trapezoidal Rule — Implementation ⭐ ⭐ ⭐ ⭐ ⭐

The uploaded material provides a Python implementation.

```
def trapezoidal_rule(f, a, b, n):
    # Create n+1 equally spaced points
    partition = [
        a + i * (b - a) / n
        for i in range(n + 1)
    ]

    # Evaluate f(x)
    vals = [f(x) for x in partition]

    # Trapezoidal rule
    integral = (
        (b - a) / (2 * n) * (vals[0] + vals[-1])
        + (b - a) / n * sum(vals[1:-1])
    )

    return integral
```

Test

Calculate:

$$\int_0^1 x^2 \, dx$$

Exact answer:

$$\frac{1}{3}$$

```
result = trapezoidal_rule(
    lambda x: x**2,
    0,
```

1,

100

)

print(result)

You should get approximately:

0.33335

As $n$increases, the approximation approaches:

$$\boxed{\frac{1}{3}}$$

The source uses this same x^2example to demonstrate the method.

---

12. Integration by Parts — ⭐⭐⭐⭐

Formula

$$\boxed{\int u\ dv = uv - \int v\ du}$$

It comes from the product rule:

$$(uv)' = u'v + uv'$$

The source describes integration by parts as the integral counterpart of the product rule.

Example

Calculate:

$$\int x \log x\ dx$$

Choose:

$$u = \log x$$

and

$$dv = x\,dx$$

Then:

$$du = \frac{1}{x}dx$$

and

$$v = \frac{x^2}{2}$$

Apply:

$$\int u\ dv = uv - \int v\ du$$

Therefore:

$$\int x \log x\ dx = \frac{x^2}{2}\log x - \int \frac{x^2}{2}\frac{1}{x}dx$$

$$= \frac{x^2}{2}\log x - \frac{1}{2}\int x\,dx$$

$$\boxed{= \frac{x^2}{2}\log x - \frac{x^2}{4} + C}$$

---

13. Integration by Substitution — ⭐ ⭐ ⭐ ⭐

Integration by substitution is essentially the **reverse of the chain rule**.

Formula

If:

$$u = g(x)$$

then:

$$du = g'(x)dx$$

and we transform the integral into terms of u.

Example

Calculate:

$$\int x \sin(x^2) dx$$

Step 1

Choose:

$$u = x^2$$

Step 2

Differentiate:

$$du = 2x\, dx$$

Therefore:

$$x\, dx = \frac{1}{2} du$$

Step 3

Substitute:

$$\int x \sin(x^2) dx = \frac{1}{2} \int \sin u \ du$$

Step 4

Integrate:

$$= -\frac{1}{2} \cos u + C$$

Step 5

Replace u:

$$\boxed{-\frac{1}{2}\cos(x^2) + C}$$

The source presents this same example.

---

14. Linearity of Integration — ⭐ ⭐ ⭐

One very useful property:

$$\boxed{\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx}$$

and:

$$\boxed{\int c\, f(x)dx = c \int f(x)dx}$$

Example

$$\int (3x^2 + 4x)dx$$

Separate:

$$3\int x^2\, dx + 4\int x\, dx$$

Therefore:

$$3\frac{x^3}{3} + 4\frac{x^2}{2}$$

$$\boxed{x^3 + 2x^2 + C}$$

The source identifies linearity as a key practical property of integration.

---

15. Integration in Machine Learning — ⭐ ⭐ ⭐ ⭐ ⭐

This is particularly important **for your ML interviews**.

Integration appears in:

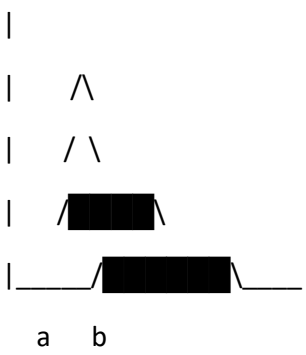
1. Probability

For a continuous random variable:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

where $f(x)$is the probability density function.

Probability density



Shaded area = probability

The source explicitly connects integration to probability and expected values.

---

2. Expected Value

For a continuous random variable:

$$\boxed{E[X] = \int_{-\infty}^{\infty} x\, f(x)dx}$$

Conceptually:

Value × probability

↓

integrate over

all possible values

↓

Expected value

---

3. Continuous Mean

Discrete average:

$$\frac{x_1 + x_2 + \cdots + x_n}{n}$$

Continuous counterpart:

$$\text{average} \sim \frac{1}{b-a}\int_a^b f\,(x)dx$$

The source explicitly describes integration as a continuous generalization of the arithmetic mean.

---

4. Probability distributions

Integration allows us to calculate:

- probability
- expectation
- variance
- moments
- cumulative probability

This is one reason calculus becomes important when moving from basic ML into probability and statistical modeling.

---

16. Riemann Sum vs Trapezoidal Rule

Very common interview comparison:

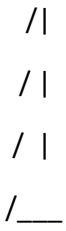
Concept	**Idea**
Riemann sum	Approximate area using small rectangles
Trapezoidal rule	Approximate area using trapezoids
Exact integration	Calculate mathematical integral

Concept	Idea
Numerical integration	Approximate integral computationally

Riemann:



Trapezoidal:



The source develops numerical integration from Riemann sums and then specializes to the trapezoidal rule.

---

17. Trapezoidal Rule vs Simpson's Rule — ⭐ ⭐ ⭐

The source also introduces Simpson's rule.

Trapezoidal

Approximates the curve with **straight lines**.

Error roughly:

$$O(n^{-2})$$

Simpson

Approximates the curve with **quadratic pieces**.

Error roughly:

$$O(n^{-4})$$

So, under the stated smoothness assumptions, increasing $n$generally makes Simpson's approximation converge faster.

Interview answer

Trapezoidal rule approximates the function piecewise linearly, whereas Simpson's rule uses piecewise quadratic approximations and therefore can achieve a faster convergence rate for sufficiently smooth functions.

---

18. The Most Important Formulas — 🔥

Memorize these.

Antiderivative

$$\boxed{\int f\,(x)dx = F(x) + C}$$

where:

$$F'(x) = f(x)$$

Fundamental Theorem

$$\boxed{\int_a^b f\,(x)dx = F(b) - F(a)}$$

Power Rule

For:

$$n \neq -1$$

$$\boxed{\int x^n\,dx = \frac{x^{n+1}}{n+1} + C}$$

Example:

$$\int x^3\,dx = \frac{x^4}{4} + C$$

Integration by Parts

$$\boxed{\int u\ dv = uv - \int v\ du}$$

Substitution

$$\boxed{u = g(x),\ du = g'(x)dx}$$

Trapezoidal Rule

$$\boxed{\int_a^b f\ (x)dx \approx \frac{b-a}{2n}\left[f(x_0) + f(x_n) + 2\sum_{i=1}^{n-1} f\ (x_i)\right]}$$

---

19. What You Should Actually Study for the Interview — 80/20

Don't spend equal time on every section.

Priority	Topic	Interview importance
🔴 P0	Integration intuition	⭐⭐⭐⭐⭐
🔴 P0	Definite vs indefinite integral	⭐⭐⭐⭐⭐
🔴 P0	Fundamental Theorem of Calculus	⭐⭐⭐⭐⭐
🔴 P0	Area / accumulation interpretation	⭐⭐⭐⭐⭐
🔴 P0	Riemann sum intuition	⭐⭐⭐⭐⭐
🔴 P0	Numerical integration	⭐⭐⭐⭐⭐
🔴 P0	Trapezoidal rule + implementation	⭐⭐⭐⭐⭐
🟠 P1	Integration by substitution	⭐⭐⭐⭐
🟠 P1	Integration by parts	⭐⭐⭐⭐
🟠 P1	Integration and probability	⭐⭐⭐⭐

Priority	Topic	Interview importance
🟡 P2	Simpson's rule	⭐⭐⭐
🟡 P2	Rigorous Riemann-integrability proofs	⭐
🟡 P2	Refinement/upper/lower sum proofs	⭐

The source itself spends substantial space establishing the rigorous Riemann theory, but for an **ML interview**, the practical 80/20 is the connection between accumulation, Riemann sums, the Fundamental Theorem, probability, and numerical integration.

---

20. Top Interview Questions

Q1. What is integration?

Answer:

Integration is the process of accumulating infinitely many small quantities. Geometrically, a definite integral represents the signed area under a curve. It can also be viewed as the inverse operation of differentiation.

Q2. What is the difference between definite and indefinite integration?

Answer:

An indefinite integral gives a family of antiderivatives and therefore contains $+C$. A definite integral has limits and produces a numerical accumulated value.

Q3. Why do we need the constant C?

Because many functions have the same derivative.

$$\frac{d}{dx}(x^2) = 2x$$
$$\frac{d}{dx}(x^2 + 10) = 2x$$

Therefore:

$$\boxed{\int 2\,x\,dx = x^2 + C}$$

Q4. What is the Fundamental Theorem of Calculus?

It connects differentiation and integration. If $F'(x) = f(x)$, then the definite integral of f from a to b equals $F(b) - F(a)$.

$$\boxed{\int_a^b f\,(x)dx = F(b) - F(a)}$$

Q5. Why is integration important in ML?

Integration is fundamental to continuous probability distributions, where probabilities and expected values are expressed as integrals. It also appears in statistical modeling and provides the mathematical basis for accumulating continuous quantities.

The uploaded source explicitly makes this connection to probability, expected value, and ML.

Q6. Why use numerical integration?

Some functions do not have convenient elementary antiderivatives. Numerical integration lets us approximate the integral computationally.

Q7. Explain the trapezoidal rule.

It divides the integration interval into smaller intervals and approximates the curve over each interval using a trapezoid. The areas of the trapezoids are then summed.

Q8. What happens when n increases in numerical integration?

Generally, the intervals become smaller, so the approximation becomes more accurate under the method's assumptions.

For the trapezoidal rule, the source gives a quadratic convergence/error rate under twice differentiability.

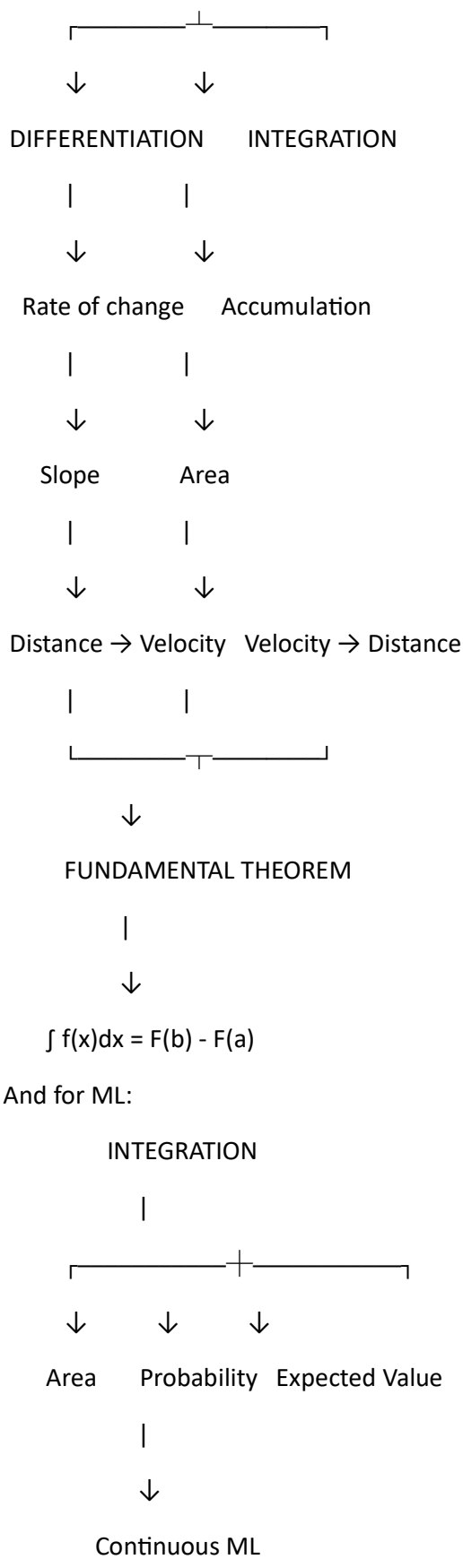
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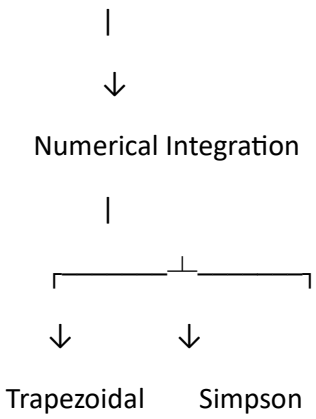
21. One Mental Model to Remember

If you remember only **one diagram**, remember this:

CALCULUS

|


DIFFERENTIATION
INTEGRATION
Rate of change
Accumulation
Slope
Area
Distance → Velocity
Velocity → Distance
FUNDAMENTAL THEOREM
∫ f(x)dx = F(b) - F(a)
And for ML:
INTEGRATION
Area
Probability
Expected Value
Continuous ML



🔥 **Final 80/20 takeaway**

For your next ML interview, make sure you can explain **without looking at notes**:

1. **What integration means intuitively**
2. **Definite vs indefinite integral**
3. **Why $+C$ exists**
4. **Integration as accumulation/area**
5. **Differentiation ↔ integration**
6. **Fundamental Theorem of Calculus**
7. **Riemann-sum intuition**
8. **Why numerical integration is needed**
9. **Trapezoidal rule**
10. **Implement trapezoidal integration in Python**
11. **Integration by substitution**
12. **Integration by parts**
13. **Why integration matters for probability and ML**

That is the highest-value 20% from this chapter for an ML-oriented interview. The source itself summarizes the same core progression: Riemann-sum definition → Newton–Leibniz formula → integration techniques → numerical methods → ML/probability connection.